

Rubidium

37
Rb

From *rubidius*, which is Latin for “deep red,” rubidium produces a red-colored flame when burned.

Discovered by the German chemists Gustav Kirchhoff and Robert Bunsen in 1861, it has a low melting point of 102.2°F (39°C).

Pure rubidium is contained in this glass to prevent it from coming into contact with air and undergoing a violent reaction.

Laboratory sample of pure rubidium in an airless vial



Fireworks

When heated in a hot flame, compounds containing group one elements produce vivid colors. Rubidium produces a red-violet flame, and rubidium nitrate is sometimes used in fireworks to give this color. It is also an ingredient in illumination flares used by the military to aid rescues and at sea as distress signals.

Cesium

55
Cs

One of the most important applications of cesium is in making atomic clocks—accurate timekeeping devices that can measure time down to a nanosecond (one thousand-millionth of a second).

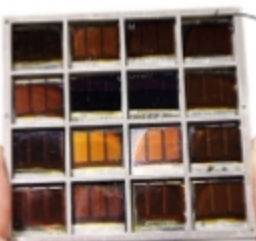
Pure cesium has a silvery-gold color.

Laboratory sample of pure cesium in an airless vial



Solar cells

Adding cesium to a solar cell boosts the conductivity of silicon in the glass, increasing the cell's efficiency. Solar cells have a wide range of applications, such as powering satellites in space.



Cesium-coated solar cell

Francium

87
Fr

This is one of the rarest elements on Earth, with only a few grams estimated to exist in the planet's crust. This scarcity meant that francium was one of the last elements to be discovered, in 1939, by the scientist Marguerite Perey. It was named after her native country—France.

Uraninite

This mineral contains tiny amounts of francium, a result of the atoms of other elements in the ore undergoing radioactive decay.

